Def Let $\underline{v}_1, \dots \underline{v}_m \in \mathbb{R}^d$. $\underline{v}_1, \dots \underline{v}_m$ are defined to be linearly indpt iff $[\underline{v}_1, \dots \underline{v}_m]$ has a pivot in each column.

(i) $\underline{v}_1 \neq \underline{0}$

(ii) col 2 is NOT a linear comb. of col 1.

$\vdots$

(m) $\underline{v}_m$ is NOT a linear comb of $\underline{v}_1 \dots \underline{v}_{m-1}$.

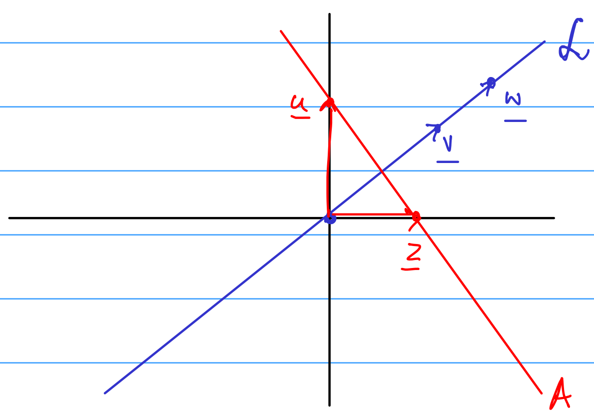
$\underline{v}_1 = \alpha \underline{v}_3 + \beta \underline{v}_4 \qquad \alpha \neq 0 \quad \beta \neq 0.$

$\underline{v}_4 = \frac{1}{\beta} \underline{v}_1 - \frac{\alpha}{\beta} \underline{v}_3.$

Def Let $\underline{v}_1, \dots \underline{v}_m \in \mathbb{R}$. $\underline{v}_1, \dots \underline{v}_m$ are defined to be linearly indpt iff $[\underline{v}_1, \dots \underline{v}_m] \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} = \underline{0}$ implies $x_1 = x_2 = \dots = x_m = 0$.

__Linear Space__

A set $\mathcal{L}$ of vectors (all with same shape) is a linear space if it is closed under linear combinations.



$\underline{v} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} \qquad \underline{w} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$

$C = \{$ all linear combinations of $\underline{v}$ and $\underline{w} \}$.

$\underline{v} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$ $\quad \underline{w} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

$C = \{$ all linear combinations of $\underline{v}$ and $\underline{w}\}$.

Does $\begin{pmatrix} 5 \\ 0 \\ 7 \end{pmatrix} \in C$?

Does $\begin{pmatrix} 2 & 1 \\ 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 7 \end{pmatrix}$ have a solution?

Elimination on $\left[\begin{array}{ccc} 2 & 1 & 5 \\ 0 & 1 & 0 \\ -1 & 0 & 7 \end{array}\right]$